

Weekly Test -6
Mathematics
Grade – XI

1. Define domain and range of a function.
2. Under what condition a function becomes bijective? Justify your answer with a suitable example.
3. Find the domain and range of the following functions:
 - (a) $\sqrt{2 - x - x^2}$
 - (b) $\sqrt{x^2 + x}$
 - (c) $\sqrt{x^2 + x - 6}$
4. Define complex number. How does it differ from a real number?
5. Define conjugate of complex number. Also, write its geometrical meaning.
6. Write any three properties of conjugate of complex numbers.
7. Solve:
 - (a) $\log_2(x^2 - 4) - \log_2(x - 2) = 5$
 - (b) $\log_x 3 = 2$
 - (c) $\log_3(x^2 + 7) = 2$
8. Find $\frac{dy}{dx}$ when:
 - (a) $y = \left(\frac{x+1}{x-1}\right)^5$
 - (b) $y = \frac{\sqrt{x}-1}{\sqrt{x}+1}$
 - (c) $y = (2x^2 + 5)^{3/2}$
 - (d) $y = 2at^2$ and $t = 3x + 5$
 - (e) $y = \frac{at^2}{t-1}$ and $x = \frac{2at}{t-1}$
9. Find $\frac{dy}{dx}$ when
 - a) $\sqrt[3]{xy} = \sqrt[3]{x} + \sqrt[3]{y}$
 - b) $(x^2 + y^2)^2 = x^2y^2$
 - c) $(x + y)^2 = x^2y^3$

